Practice workbook · Volume of revolution about the y-axis

Rewriting the curve as x^2 in terms of y and finding pi times the integral of x^2 dy between y-limits.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-24

1. Question 4 · part b · 4 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part b only.

4

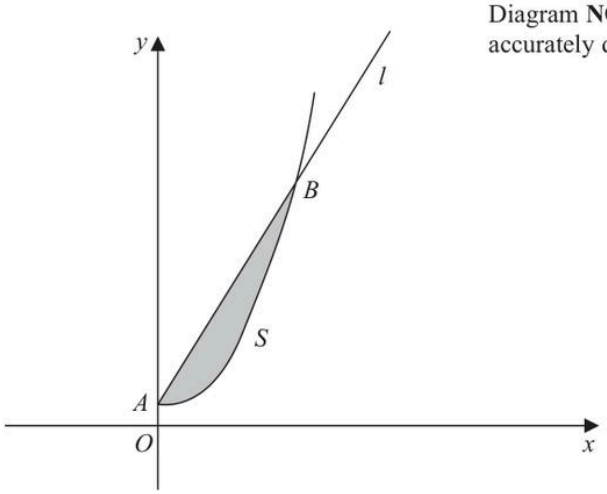


Diagram **NOT** accurately drawn

Figure 2

The curve S with equation $y = \frac{x^2}{4} + 2$ where $x \geqslant 0$ and the line l with equation $2y - x - 4 = 0$ where $x \geqslant 0$ intersect at the points A and B, as shown in Figure 2.

(a) (i) Show that the coordinates of point A are $(0, 2)$

(ii) Find the coordinates of the point B

(4)

The finite region bounded by S and l, shown shaded in Figure 2, is rotated through 2π radians about the y-axis.

(b) Use algebraic integration to find the volume of the solid generated. Give your answer in terms of π

(4)

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1. Question 4 · part b · 4 marks

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Question 4 continued

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1. Question 4 · part b · 4 marks

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Worked-example retry · Attempt part b only.

Question 4 continued

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1. Question 4 · part b · 4 marks

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Worked-example retry · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 8 marks)

1. Question 4 · part b · 4 marks · Working and answer

Working and answer

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2. Question 9 · part whole · 9 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

9 The finite region R enclosed by the y-axis, the straight line with equation $y + 2x = 13$ and the curve with equation $y = x^2 - 2$, is defined for points with coordinates (x, y) with $x \geqslant 0$

The region R is rotated through 360° about the y-axis.

Use algebraic integration to find the volume of the solid generated.
Give your answer in terms of π.

(9)

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2. Question 9 · part whole · 9 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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2. Question 9 · part whole · 9 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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2. Question 9 · part whole · 9 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

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(Total for Question 9 is 9 marks)

2. Question 9 · part whole · 9 marks · Working and answer

Working and answer

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3. Question 7 · part whole · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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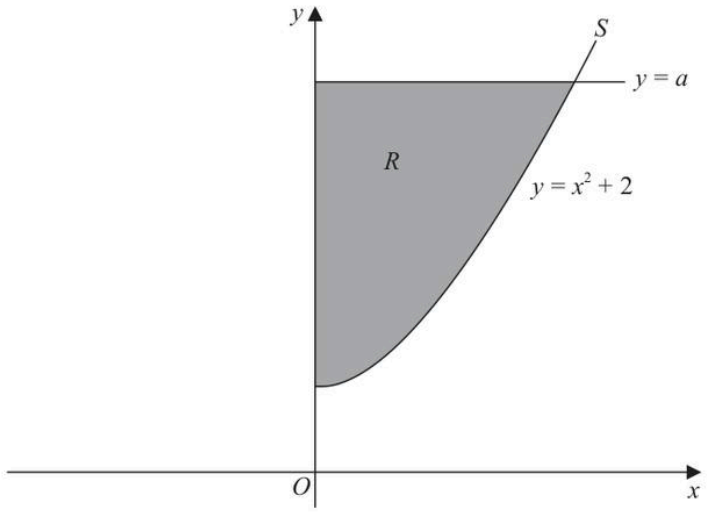


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve S with equation $y = x^2 + 2$

The finite region R, shown shaded in Figure 3, is bounded by S, the y-axis and the line with equation $y = a$ where $a > 2$

The region R is rotated through 360° about the y-axis to generate a solid with volume 18π

Use algebraic integration to find the value of a

(8)

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3. Question 7 · part whole · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 7 continued

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3. Question 7 · part whole · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 7 continued

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3. Question 7 · part whole · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 7 continued

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(Total for Question 7 is 8 marks)

3. Question 7 · part whole · 8 marks · Working and answer

Working and answer

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4. Question 6 · part b · 7 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

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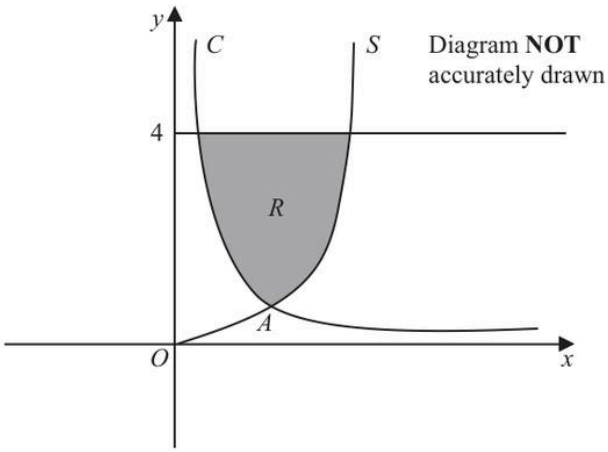


Figure 3

Figure 3 shows part of the curve C with equation $y = \frac{1}{4x}$, $x > 0$ and part of the curve S with equation $y = 2x^2$, $x \geqslant 0$

The curve C and the curve S intersect at the point A

(a) Find the coordinates of point A

(3)

The finite region R, shown shaded in Figure 3, bounded by the curve C, the curve S and the straight line $y = 4$ is rotated through 360° about the y-axis.

(b) Find, using algebraic integration, the exact volume of the solid formed.

(7)

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4. Question 6 · part b · 7 marks

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Further practice · Attempt part b only.

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Question 6 continued

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(Total for Question 6 is 10 marks)

4. Question 6 · part b · 7 marks · Working and answer

Working and answer

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